

Fairness After the Decision: Group Fairness Under Selective Human Contestation

RANJAN VEERABHADRASWAMY, Vellore Institute of Technology, Andhra Pradesh, India

AJITH JUBILSON EMERSON*, Vellore Institute of Technology, Andhra Pradesh, India

Background: Group-fairness analyses often stop at an algorithm’s initial decision even when affected people may enter a selective human review process. **Objectives:** We characterize how claimant-initiated contestation changes standard group-fairness quantities and what ordinary appeal logs identify. **Methods:** A label- and group-specific post-decision operator separates appeal access from review quality. We derive exact preservation and signed-gap conditions, analyze heterogeneous-cost access and budgeted assistance, and obtain appeal-log-specific sharp bounds. Synthetic checks and 600 paired semisynthetic Adult/German Credit runs evaluate the formal predictions. **Results:** Adult supports signed movement and absolute equal-opportunity-gap amplification in all 12 unequal-access model-scenario cells. On German Credit, signed movement is resolved in 7 of 12 cells and in none of the four random-forest cells; no cell supports broad absolute-gap amplification. **Conclusions:** Selective review can create, reinforce, attenuate, cross, or reverse disparity. Empirical validation is partial rather than universal, and auditing must cover the initial decision, access, review, and final action.

1 Introduction

A denied loan, benefit, job, or service is not always the final institutional action. A person may submit evidence, navigate an appeal process, or obtain human review. Entry into that process is rarely random: time, information, documentation, language, legal help, and expected benefit all affect who contests. Consequently, two groups can face the same formal appeal rule but different effective opportunities to correct an error.

This observation creates a distinct fairness problem. Classifier metrics describe the initial decision, whereas people experience the final decision after review. A correction channel can improve aggregate accuracy while redistributing that improvement unequally. Conversely, differential review can sometimes reduce an existing absolute disparity. The relevant object is thus not an informal “right to appeal,” but a stochastic post-decision transformation whose transition probabilities depend on group, latent label, initial outcome, and policy.

We study a binary, one-shot model in which an individual with an adverse initial decision may appeal and review may reverse that decision. Let α_g^+ be the probability that a false denial in group g is appealed and ρ_g^+ its conditional correction probability; let α_g^- and ρ_g^- analogously describe an appealed correct denial that is incorrectly reversed. The effective transition rates are $\kappa_g^+ = \alpha_g^+ \rho_g^+$ and $\kappa_g^- = \alpha_g^- \rho_g^-$. This factorization separates access from review quality while making clear that final fairness depends on their product.

Our contributions are deliberately narrow: (i) exact fairness transformation and preservation characterization for a post-decision contestation operator; (ii) signed disparity dynamics, including creation, amplification, attenuation, crossing, and reversal; (iii) group-fairness consequences of endogenous participation under heterogeneous costs and formally symmetric policies; (iv) appeal-log-specific non-identification and sharp final-FNR and disparity bounds; and (v) resource-constrained assistance within the same mechanism. We do not claim generic novelty for contestability, procedural fairness, recourse burdens or costs, selective labels, partial identification, random auditing, convex optimization, or knapsack structure.

The empirical evidence is intentionally qualified. Synthetic experiments verify the formal results. Semisynthetic experiments use real prediction tasks but simulated contestation, not observed human appeals. Adult robustly

*Corresponding author.

Authors’ Contact Information: Ranjan Veerabhadraswamy, School of Computer Science and Engineering, Vellore Institute of Technology, Andhra Pradesh, Amaravati, Andhra Pradesh 522241, India, ranjanvswamyjnv2005@gmail.com; Ajith Jubilson Emerson, School of Computer Science and Engineering, Vellore Institute of Technology, Andhra Pradesh, Amaravati, Andhra Pradesh 522241, India, ajith.jubilson@vitap.ac.in.

shows the proposed mechanism; the smaller German Credit data support the directional theory for two models but not a model-robust absolute-gap claim.

2 Related work and positioning

Contestability and correctability research motivates rights, evidence, and infrastructure for challenging automated decisions (Cobbe et al. 2021; Freiesleben et al. 2026; Kaminski and Urban 2021; Lyons, Miller, et al. 2022; Lyons, Velloso, et al. 2021). Perdana’s “correctability gap” is a useful motivation: conventional fairness metrics do not show whether affected people can appeal, submit evidence, or obtain human review (Perdana 2026). We formalize one such correction channel rather than treating correctability itself as new.

Procedural fairness studies the fairness of the decision-making process. Wang et al. formalize individual and group procedural fairness using feature-attribution characterizations of the model process (Wang et al. 2026). Our object begins after the model has issued D_0 : self-selected access to human review and the resulting D_1 . Relatedly, human discretion can change fairness in machine-assisted decisions (Angelova et al. 2023; Gillis et al. 2022); here review is claimant-initiated rather than discretion applied across all cases.

Differential parity compares two sets of decisions and asks whether their difference is independent of a sensitive attribute, including human and model comparisons and bridge estimation across different data (Yu et al. 2026). We do not propose another relative-fairness metric. We specify the transition $D_0 \rightarrow \text{appeal} \rightarrow \text{human review} \rightarrow D_1$, make participation endogenous, and derive its transformation of standard fairness quantities, identification properties, and interventions.

Recourse work shows that feasible actions, costs, and effectiveness vary across groups (Kavouras et al. 2023; Kügelgen et al. 2022). Nitchals separately analyzes heterogeneous private appeal costs, procedural pricing, correction, resource use, deterrence, and an interior optimal price (Nitchals 2026). We make no novelty claim for heterogeneous costs or resource-aware procedure design. Our focus is protected-group fairness, preservation of standard criteria, effective group-specific correction, appeal-log identification, and fairness-aware assistance.

Selective-label research studies learning and fairness when outcomes are observed only for selected cases (De-Arteaga et al. 2018; Chen et al. 2025; Coston et al. 2021; Kilbertus et al. 2020; Wei 2021). Valdrighi et al. address long-term fairness with selectively observed labels and sequential policies (Valdrighi et al. 2026); Liu et al. study identification and inference for fairness–accuracy frontiers (Liu et al. 2026). We do not claim generic partial-identification novelty. Our missingness comes from self-selection into a post-decision correction channel, and we derive appeal-specific sharp bounds for final FNR and associated group disparity.

Thus existing work separately studies contestability, procedural fairness, relative decision-set fairness, human discretion, recourse burdens, appeal economics, and selective-label identification. We instead fix a specific mechanism—an adverse AI decision may enter a self-selected human correction channel—and ask how it transforms standard group-fairness properties and what ordinary appeal records identify.

3 Model

Let $G \in \mathcal{G}$ denote group membership, $Y \in \{0, 1\}$ the target label, $D_0 \in \{0, 1\}$ the initial decision, and $D_1 \in \{0, 1\}$ the final decision. Decision 0 is adverse. Write

$$t_g = \mathbb{P}(D_0 = 1 \mid Y = 1, G = g), \quad f_g = \mathbb{P}(D_0 = 1 \mid Y = 0, G = g),$$

with $\text{FNR}_g = 1 - t_g$ and $\text{TNR}_g = 1 - f_g$. Let A indicate appeal and R indicate reversal after review. In the core one-sided model, only $D_0 = 0$ cases can transition to $D_1 = 1$. Define

$$\alpha_g^y = \mathbb{P}(A = 1 \mid D_0 = 0, Y = y, G = g), \quad \rho_g^y = \mathbb{P}(R = 1 \mid A = 1, D_0 = 0, Y = y, G = g),$$

and $\kappa_g^y = \alpha_g^y \rho_g^y$. The superscripts $+$ and $-$ refer to $y = 1$ and $y = 0$, respectively.

This model assumes fixed groups and outcomes, stable review behavior within a policy environment, and a one-shot appeal. It does not require people to know whether the initial decision is erroneous. A general Markov kernel can allow both directions of revision, but the one-sided form yields transparent fairness identities.

4 Contestation transforms fairness

PROPOSITION 1 (ONE-SIDED TRANSFORMATION). *For every group g ,*

$$\text{TPR}'_g = t_g + (1 - t_g)\kappa_g^+, \quad \text{FNR}'_g = (1 - t_g)(1 - \kappa_g^+), \quad (1)$$

$$\text{FPR}'_g = f_g + (1 - f_g)\kappa_g^-, \quad \text{TNR}'_g = (1 - f_g)(1 - \kappa_g^-). \quad (2)$$

The proof partitions each label stratum by its initial decision and applies the relevant transition probability. More generally, if p_g is a row vector over decisions and K_g^y is a label-specific stochastic kernel, then $p'_g = p_g K_g^y$. The closed forms above are its adverse-to-favorable specialization.

For two groups a, b , define the signed equal-opportunity gap $\Delta_{EO} = t_a - t_b$. The exact post-contestation gap is

$$\Delta'_{EO} = \Delta_{EO} + (1 - t_a)\kappa_a^+ - (1 - t_b)\kappa_b^+. \quad (3)$$

Thus equal opportunity is preserved if and only if the two correction increments are equal. When $t_a = t_b = t$, contestation creates the gap $(1 - t)(\kappa_a^+ - \kappa_b^+)$. When a gap already exists, however, the increment can reinforce, attenuate, cross, or reverse it. Equation (3), not monotone absolute amplification, is the general prediction.

The false-positive signed gap has the parallel form

$$\Delta'_{FPR} = f_a - f_b + (1 - f_a)\kappa_a^- - (1 - f_b)\kappa_b^-.$$

Equalized odds requires both increment equalities. Demographic parity is different. If $\pi_g = \mathbb{P}(Y = 1 \mid G = g)$, its within-group positive-rate increment is

$$\pi_g \text{FNR}_g \kappa_g^+ + (1 - \pi_g) \text{TNR}_g \kappa_g^-. \quad (4)$$

Equal transition rates therefore need not preserve demographic parity because groups may contain different masses and compositions of appealable cases; unequal transitions can also offset those differences.

5 Formal symmetry and endogenous access

Suppose an eligible individual appeals when perceived benefit v exceeds net cost $C_g - s_g$, where s_g is assistance. If C_g has conditional CDF F_g , then

$$\alpha_g(s_g) = F_g(v + s_g). \quad (5)$$

The same posted procedure or subsidy does not imply the same appeal probability when cost distributions differ. With common review success ρ , effective correction is $\rho F_g(v + s_g)$.

If $F_a(x) > F_b(x)$ throughout the policy-relevant support and assistance must be group blind, $s_a = s_b$, then equal effective access is unattainable in the interior unless a boundary saturates both groups or another component of the mechanism offsets the ordering. This is a conditional impossibility result, not a claim that equality is impossible under all policy designs. Group-specific assistance, changes to documentation requirements, review-quality interventions, or saturation provide escape routes.

6 Allocating limited assistance

Let $x_g \geq 0$ be assistance to group g , n_g its eligible population, and B a budget. A planner may minimize aggregate final error plus a fairness penalty,

$$\min_x \quad L(x) + \lambda \Phi(x) \quad \text{s.t.} \quad \sum_g n_g x_g \leq B, \quad (6)$$

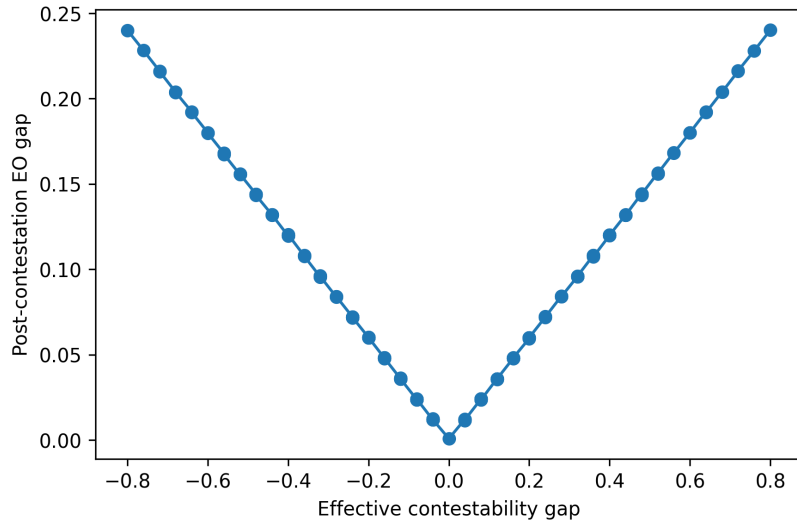


Fig. 1. Synthetic verification of the relationship between effective contestability differences and the post-contestation equal-opportunity gap.

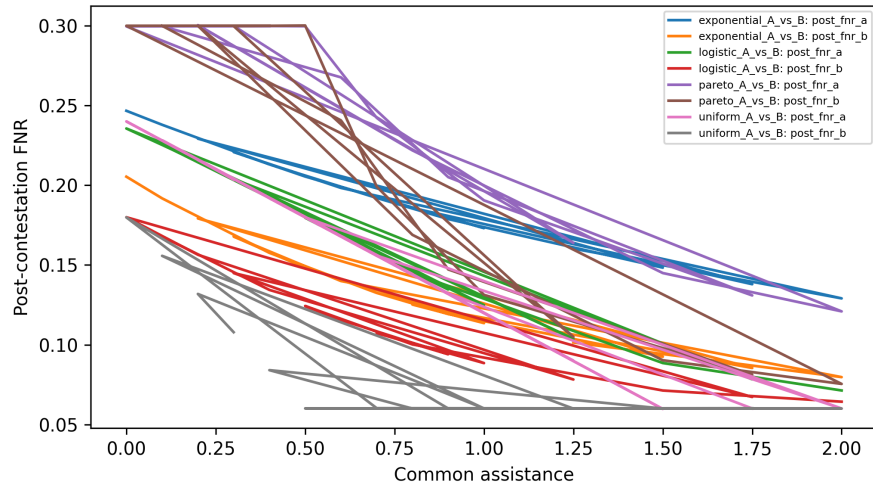


Fig. 2. A formally equal assistance policy can produce unequal effective access when groups face different appeal-cost distributions.

where appeal access follows Equation (5). For differentiable, convex separable-welfare instances satisfying Slater's condition, with a binding budget and an interior allocation, the KKT conditions equate weighted marginal harm reduction per unit resource among assisted groups, adjusted by the shadow price of budget. A fairness-aware optimum can therefore prioritize a group whose marginal disparity reduction is large even when a purely accuracy-maximizing rule would not.

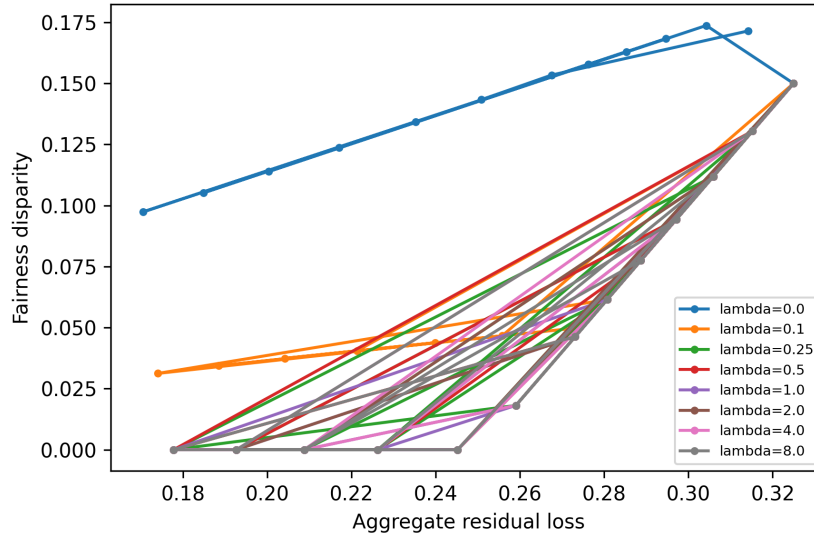


Fig. 3. The experimental fairness–utility frontier under limited assistance. The curve makes the normative trade-off explicit.

For indivisible interventions, the problem contains 0–1 knapsack as a special case; this classical reduction motivates exact dynamic programming or approximation rather than a claim of new complexity theory. Our experiments compare the fairness-aware rule with uniform, random, and proportional baselines and expose the parameter λ rather than silently fixing a normative objective.

7 What appeal logs identify

An ordinary log observes G, D_0, A and the review outcome for appellants, but typically not a valid Y for non-appellants. Let m_g be the adverse-decision mass, $q_g = \mathbb{P}(A = 1 \mid D_0 = 0, G = g)$, and let r_g be the observed positive-label share among appellants. The latent positive share among non-appellants, u_g , can vary over $[0, 1]$ without changing the appeal-log distribution. Hence the positive mass among adverse cases lies sharply in

$$m_g\{q_g r_g + (1 - q_g)u_g\}, \quad u_g \in [0, 1]. \quad (7)$$

Combining this interval with the observed positive mass among initially favorable cases yields sharp bounds on the group FNR. Distinct worlds can therefore have identical appeal logs but materially different oracle error rates and fairness conclusions.

Representative random audits of adverse non-appellants identify u_g under valid labels and known inclusion probabilities. Inverse-probability estimation is then consistent, and uncertainty contracts with the effective audit sample size. Random auditing is a supporting design result familiar from missing-data and selective-label settings; its importance here is that it closes the precise gap left by self-selected appeal logs.

8 Experiments

8.1 Design

The evaluation has six predeclared gates. Unit and smoke tests precede the scientific runs. Synthetic experiments then test algebraic agreement, interior-grid directional predictions, assistance allocation, observational equivalence, sharp bounds, and audit-width contraction. The final gate uses Adult (Becker and Kohavi 1996) and

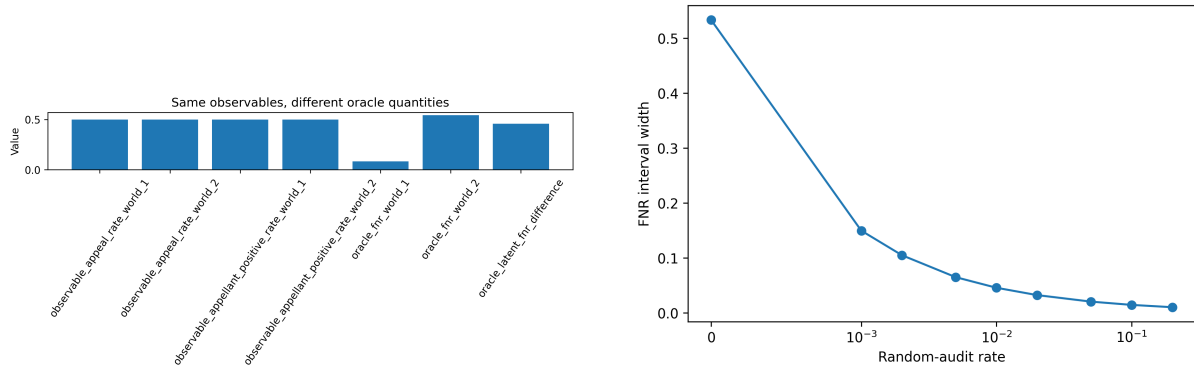


Fig. 4. Left: observationally equivalent appeal logs can conceal different oracle FNRs. Right: representative auditing contracts the identified interval.

German Credit (Hofmann 1994) as real classification tasks with a semisynthetic post-decision mechanism. Both are public UCI datasets distributed under CC BY 4.0.

For each real dataset we fit logistic regression, gradient boosting, and random forest models. Train, validation, and test partitions are separated; group thresholds are calibrated on validation data and frozen before test evaluation. Five contestation regimes are fixed in advance: equal access and moderate or strong advantage for either group. Twenty paired seeds (1729–1748) yield $2 \times 3 \times 5 \times 20 = 600$ real-data runs. We report two-sided 95% Student-*t* intervals for paired seed-level pre/post changes. These data do not contain observed appeals, costs, or review behavior; only the prediction task is empirical.

The complete pipeline contains 820 standardized seed-level CSV files: 220 synthetic/theorem files and 600 real-data files. All commands returned zero, and validation completed without errors or warnings. The downloaded Kaggle artifact is retained with a SHA-256 digest and an immutable extracted copy.

8.2 Dataset and protocol data card

Both prediction datasets are from the UCI Machine Learning Repository and are distributed under CC BY 4.0. Adult combines the 32,561 training and 16,281 test records supplied by UCI; all 48,842 records remain after deterministic schema conversion. The target is income $> \$50,000$ (favorable = 1; $\leq \$50,000$ unfavorable = 0), and the protected attribute is sex coded exactly as Female/Male. German Credit contains 1,000 raw and 1,000 prepared records. UCI credit-risk label 1 (good) maps to favorable 1 and label 2 (bad) to 0. Personal-status/sex codes A92 and A95 map to Female; A91, A93, and A94 map to Male. In both datasets the group column is removed from classifier inputs. German’s source personal-status/sex field is also removed.

Adult question marks are parsed as missing; German conversion introduces no missing-value rule. In the modeling pipeline, numeric missing values are median-imputed and standardized; categorical missing values are most-frequent-imputed and one-hot encoded with unknown levels ignored. For each seed, joint label-group stratification creates 60/20/20 percent train/validation/test partitions: test uses seed s and validation uses $s + 1$. Logistic regression, histogram gradient boosting, and random forest are fit on training data. On validation data, each group’s threshold is selected from $\{0.01, \dots, 0.99\}$ to match the overall positive-class TPR at threshold 0.5 as closely as possible; thresholds are then frozen on test.

The paired seed range is 1729–1748. The five preregistered regimes set the Female/Male values of α^+ as follows: equal (.50, .50); moderate female (.65, .35); strong female (.75, .25); moderate male (.35, .65); and strong male (.25, .75). Every regime uses Female/Male values $\alpha^- = (.25, .25)$, $\rho^+ = (.80, .80)$, and $\rho^- = (.05, .05)$.

Table 1. Summary of unequal-access E6 cells. “Signed resolved” means the paired 95% interval excludes zero in the direction specified by the access advantage.

Dataset	Cells	Signed resolved	Absolute increase	Accuracy increase
Adult	12	12	12	12
German Credit	12	7	0	12

Dataset features, targets, and initial predictions are empirical; appeal, review, and final-decision variables are semisynthetic.

8.3 Synthetic results

The synthetic gates pass. Across the EXP-01 grid, theoretical signed gaps span -0.24 to 0.24 and empirical means have absolute error between 0.000441 and 0.001621 . EXP-02 covers signed gaps from -0.15 to 0.15 with mean absolute error from 0.000640 to 0.001522 . The heterogeneous- cost experiment has mean absolute errors from 0.000379 to 0.001324 .

At interior budgets, the fairness-aware allocation produces a nontrivial trade-off between the absolute equal-opportunity gap and aggregate loss. It substantially reduces the absolute gap relative to uniform, random, and proportional baselines. In the identification suite, four pairs of observationally equivalent worlds have oracle FNR differences from $1/3$ to $2/3$. Aggregate interval width falls from 0.1496 at audit rate 0.001 to 0.0102 at rate 0.20 , with reported coverage 1.0 on the evaluated grid.

8.4 Semisynthetic real-data results

Table 1 gives the predeclared E6 decision. On Adult, average pre-contestation absolute equal-opportunity gaps range from 0.031 to 0.039 . Every one of the 12 unequal-access model–scenario cells increases the absolute gap, with all paired 95% intervals excluding zero. Mean increases range from 0.0442 to 0.1513 . All 12 signed shifts point toward the group with greater false-denial access and exclude zero.

German Credit behaves differently. Average initial gaps are 0.0275 for random forest, 0.0742 for gradient boosting, and 0.0945 for logistic regression. The expected signed movement is resolved in 7 of 12 unequal-access cells: $4/4$ logistic, $3/4$ gradient boosting, and $0/4$ random forest. Signed mean changes range from -0.0700 to $+0.0443$. No German cell has a paired interval supporting an increase in the absolute gap. The signed disparity often attenuates, crosses zero, or reverses, consistent with Equation (3), while small test strata produce noisier calibration and inference.

Overall accuracy improves with a positive paired interval in all 30 dataset–model–scenario cells. The German null is therefore not a failed execution: review remains mostly corrective, but its fairness effect is neither broad nor model robust. We classify E6 as a *partial pass*.

9 Discussion and limitations

The main conceptual implication is that classifier fairness and final-decision fairness are different estimands. A system can become more accurate for every group yet distribute corrections unequally. Audits should record initial and final decisions, appeal eligibility, attempted and completed appeals, review outcomes, processing time, assistance, and representative labels among non-appellants. Reporting appeal success only among those who enter the process is insufficient.

The theory does not rank fairness definitions or resolve normative trade-offs. It assumes binary labels and decisions, a one-shot channel, stable review behavior, and fixed groups. It omits queues, deadlines, repeated

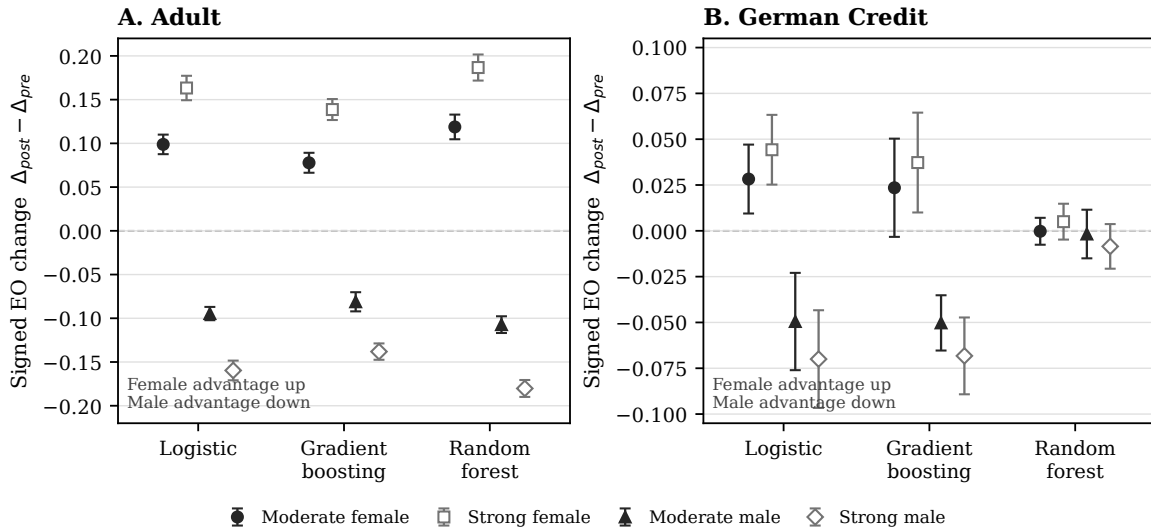


Fig. 5. Paired change in the signed equal-opportunity disparity $\Delta_{\text{post}} - \Delta_{\text{pre}}$ under the four unequal-access regimes; bars are two-sided 95% paired Student- t intervals across 20 seeds. The sign is Female minus Male, so female-advantage regimes move upward and male-advantage regimes downward. Adult resolves all 12 predicted directions; German Credit resolves 7 of 12, and none for random forest. Absolute-gap changes are summarized in Table 1: 12/12 positive on Adult and 0/12 positive on German Credit.

appeals, retaliation, classifier retraining, strategic institutions, and causal effects of protected-group membership. Reviewer accuracy can depend on case mix, so expanding access may change ρ rather than merely α . Assistance may alter perceived benefit or evidentiary quality as well as cost.

Labels may be delayed, disputed, or affected by the decision. Random audits identify non-appellant outcomes only when selection is representative and audit labels are valid. Group-level equality can conceal intersectional and within-group inequity. The real-data component is semisynthetic and contains no institutional estimates of appeal behavior or costs. German Credit, in particular, does not support a universal absolute-amplification claim.

The supplied novelty audit also limits the claim. Contestability, reviewability, procedural fairness, relative decision-set fairness, fair recourse, heterogeneous appeal costs, human discretion, selective labels, partial identification, random auditing, convex KKT conditions, and knapsack structure are established topics. The contribution is the specified claimant-initiated correction mechanism and its exact fairness, identification, and intervention consequences; no priority claim is made.

10 Conclusion

Post-decision review is not automatically a fairness safeguard. Within the specified claimant-initiated mechanism, exact transformation identities state when equal opportunity, false-positive parity, equalized odds, or demographic parity survive review. The signed correction term can reinforce, attenuate, cross, or reverse an existing disparity. Endogenous participation explains why formal symmetry can fail to produce effective symmetry; the same model supports constrained assistance analysis and appeal-log-specific sharp bounds. The evidence marks the empirical boundary honestly: consistent Adult effects, partial directional support on German Credit, and no universal absolute-gap amplification result.

Reproducibility statement

The final audited Kaggle run is 352782151 (CPU; Internet disabled), executed 2026-09-25 18:51:07–19:12:56 UTC. Its downloaded archive has SHA-256

AE30CA19BF912DB8FEB1BE33E263C994
F73ED3C2556E6EFF6D6EB69747BCBF1A.

The analysis validates the exact factorial before computing paired intervals. Configuration, seeds, standardized outputs, figures, gate decisions, and an output-inventory regression test accompany the manuscript.

Code, configurations, and reproducibility artifacts are available at <https://github.com/Ranjan-V/JAIR-Fairness-After-the-Decision>. The source code is released under the MIT License, and author-generated research artifacts are released under CC BY 4.0. Third-party datasets retain their original source terms and are not redistributed in the repository.

A fresh clean-environment reproduction regenerated all 220 synthetic/theorem outputs. All 220 matched the archived numerical content at absolute and relative tolerance 10^{-6} ; the maximum absolute numerical difference was 5.39×10^{-8} . This is a scientific-content reproduction, not a claim of byte equality. Cross-platform line endings, serialized parameter keys, and final optimizer digits differ across the audited Linux and Windows environments; these differences are documented in the reproduction audit.

A Technical Appendix: Proofs and Boundary Cases

This appendix states the exact scope of every named mathematical result used by the paper. Conditional probabilities are asserted only when their conditioning events have positive probability. All one-sided results assume that only an adverse initial decision can change and that a non-appellant retains the initial decision.

A.1 Operators and confusion-rate transformations

Theorem 1 (linear operator preservation criterion). Let V be a finite-dimensional vector space of feasible signed perturbations, let $T_K : V \rightarrow V$ be the linear map induced by a fixed collection of stochastic kernels, and let $L : V \rightarrow W$ encode a homogeneous linear fairness constraint. The following are equivalent:

- (1) $Lv = 0$ implies $LT_K v = 0$ for every $v \in V$;
- (2) $\ker L \subseteq \ker(LT_K)$;
- (3) there is a linear map $M : \text{im } L \rightarrow W$ such that $LT_K = ML$ on V .

PROOF. The first two conditions are identical. Under the second, define $M(Lv) = LT_K v$. If $Lv = Lw$, then $v - w \in \ker L$, so $LT_K(v - w) = 0$ and the definition does not depend on the representative. It is linear by construction. Conversely, $LT_K = ML$ immediately implies $LT_K v = 0$ whenever $Lv = 0$. $\square$

For probability models, V must be the span of feasible signed differences. An assertion made only on a thin or boundary-only subset of probability vectors does not automatically extend to the full algebraic kernel. Affine constraints must be homogenized with a constant coordinate.

Proposition 1 (one-sided confusion-rate transformation). For every group g with defined label-conditional rates,

$$\begin{aligned} \text{TPR}'_g &= \text{TPR}_g + \text{FNR}_g \kappa_g^+, & \text{FNR}'_g &= \text{FNR}_g (1 - \kappa_g^+), \\ \text{FPR}'_g &= \text{FPR}_g + \text{TNR}_g \kappa_g^-, & \text{TNR}'_g &= \text{TNR}_g (1 - \kappa_g^-). \end{aligned}$$

PROOF. Conditional on $Y = 1, G = g$, a final positive is either an initial positive or an initial negative followed by appeal and reversal. These events are disjoint, and the latter has probability $\text{FNR}_g \alpha_g^+ \rho_g^+ = \text{FNR}_g \kappa_g^+$. Complementation gives the FNR identity. Repeating the argument in the $Y = 0$ stratum gives FPR and TNR. $\square$

Corollary 1 (positive-decision rate). With $\pi_g = \mathbb{P}(Y = 1 \mid G = g)$ and $r_g = \mathbb{P}(D_0 = 1 \mid G = g)$,

$$r'_g = r_g + \pi_g \text{FNR}_g \kappa_g^+ + (1 - \pi_g) \text{TNR}_g \kappa_g^-.$$

PROOF. Condition on Y and substitute Proposition 1. $\square$

Proposition 2 (two-sided transformation). If $q_g^y = \mathbb{P}(D_1 = 0 \mid D_0 = 1, Y = y, G = g)$, then

$$\text{TPR}'_g = t_g(1 - q_g^1) + (1 - t_g)\kappa_g^+, \quad \text{FPR}'_g = f_g(1 - q_g^0) + (1 - f_g)\kappa_g^-.$$

PROOF. Partition on D_0 within each label stratum and apply the two rows of the transition kernel. $\square$

A.2 Fairness preservation and signed disparity

Theorem 2 (equal-opportunity preservation). If $t_A = t_B = t$, then post-contestation equal opportunity holds if and only if $(1 - t)(\kappa_A^+ - \kappa_B^+) = 0$. When $t < 1$, this is equivalent to $\kappa_A^+ = \kappa_B^+$; when $t = 1$, it is automatic.

PROOF. Subtract the two TPR identities in Proposition 1. $\square$

Theorem 3 (FPR-parity preservation). If $f_A = f_B = f$, then post-contestation FPR parity holds if and only if $(1 - f)(\kappa_A^- - \kappa_B^-) = 0$.

PROOF. Subtract the two FPR identities in Proposition 1. $\square$

Corollary 2 (equalized odds). Under initial equalized odds, final equalized odds holds exactly when both conditions in Theorems 2 and 3 hold.

PROOF. Equalized odds is the conjunction of equal TPR and equal FPR. $\square$

Theorem 4 (demographic-parity preservation). If $r_A = r_B$ initially, demographic parity is preserved exactly when

$$\pi_A \text{FNR}_A \kappa_A^+ + (1 - \pi_A) \text{TNR}_A \kappa_A^- = \pi_B \text{FNR}_B \kappa_B^+ + (1 - \pi_B) \text{TNR}_B \kappa_B^-.$$

PROOF. Subtract the two identities in Corollary 1 and use $r_A - r_B = 0$. $\square$

Proposition 3 (universal one-sided preservation). At a fixed common initial TPR $t < 1$, a pair of one-sided operators preserves equal opportunity for every admissible group distribution if and only if the two $Y = 1$ adverse-to-positive transition probabilities agree. The analogous claim holds for FPR at a common $f < 1$.

PROOF. Sufficiency follows from Theorem 2. Necessity follows because the multiplier $1 - t$ (or $1 - f$) is strictly positive. $\square$

Theorem 5 (exact signed-gap dynamics). Let $d = t_A - t_B$ and $c = (1 - t_A)\kappa_A^+ - (1 - t_B)\kappa_B^+$. Then $d' = d + c$ and

$$|d'| > |d| \iff c(2d + c) > 0,$$

$$|d'| < |d| \iff c(2d + c) < 0,$$

$$|d'| = |d| \iff c = 0 \text{ or } c = -2d.$$

Direction reverses exactly when $d(d + c) < 0$, and final equality holds exactly when $c = -d$.

PROOF. The identity $d' = d + c$ follows from Proposition 1. Since absolute gaps are nonnegative, $|d + c|^2 - |d|^2 = c(2d + c)$ has the same sign as their difference. The remaining claims follow directly from the signs of $d, d + c$ and from $d + c = 0$. $\square$

Theorem 6 (tight attainable bounds). For unrestricted independent $\kappa_A^+, \kappa_B^+ \in [0, 1]$,

$$d' \in [t_A - 1, 1 - t_B].$$

The minimum attainable absolute gap is zero when this interval contains zero and otherwise is the distance from the interval to zero; the maximum is the larger absolute endpoint.

PROOF. Writing $a = 1 - t_A$ and $b = 1 - t_B$, the correction $a\kappa_A^+ - b\kappa_B^+$ ranges continuously over $[-b, a]$. Adding d gives the displayed interval. The absolute-value extrema follow from elementary interval geometry and are attained at a corner or a feasible root. $\square$

Corollary 3 (initially fair case). If $t_A = t_B = t$, then $|d'| = (1 - t)|\kappa_A^+ - \kappa_B^+| \leq 1 - t$, and the bound is tight.

A.3 Endogenous access and the conditional impossibility

Proposition 4 (threshold appeal response). If perceived value is deterministic at v_g^y and the conditional cost CDF is F_g^y , then $\alpha_g^y(u) = F_g^y(v_g^y + u)$. For random value V ,

$$\alpha_g^y(u) = \mathbb{E}[F_{C|V,g,y}(V + u \mid V, G = g, Y = y, D_0 = 0)].$$

PROOF. The appeal event is $\{C \leq V + u\}$. Condition on the declared stratum and, in the random-value case, condition once more on V . $\square$

Proposition 5. Comparative statics. The appeal response is nondecreasing and right-continuous in assistance. If the conditional cost law has density $f_{C|V,g,y}$ and differentiation under the expectation is valid, then

$$(\alpha_g^y)'(u) = \mathbb{E}[f_{C|V,g,y}(V + u \mid V, g, y)] \geq 0.$$

PROOF. The indicator of $C \leq V + u$ is pointwise nondecreasing in u , and its expectation is a shifted CDF mixture. Right-continuity follows from CDF right-continuity and bounded convergence. Under the stated regularity, differentiation under the expectation yields the density formula. $\square$

Theorem 7 (formal symmetry versus effective equality). Under common assistance u , deterministic perceived value v , and common review success $\rho > 0$,

$$\kappa_A^+(u) - \kappa_B^+(u) = \rho\{F_A^+(v + u) - F_B^+(v + u)\}.$$

Thus effective correction parity holds exactly when the two conditional CDFs agree at the operative threshold.

PROOF. Apply Proposition 4 and multiply each appeal probability by the common review-success rate. $\square$

Theorem 8 (group-blind policy impossibility). Assume initial $t_A = t_B = t < 1$, group-blind u in a feasible set U , common v , common $\rho > 0$, and $F_A^+(v + u) < F_B^+(v + u)$ for every $u \in U$. No feasible group-blind policy preserves equal opportunity.

PROOF. Theorem 7 gives $\kappa_A^+(u) < \kappa_B^+(u)$. Theorem 2 then gives

$$t'_A - t'_B = (1 - t)\rho\{F_A^+(v + u) - F_B^+(v + u)\} < 0.$$

The strictness assumptions are essential: saturation, a crossing point, zero review success, a perfect initial classifier, unequal review quality, or group-specific assistance can each invalidate the conclusion. $\square$

A.4 Limited-resource allocation

Define $R_g(u_g) = \text{FNR}_g[1 - \rho_g^+ \alpha_g^+(u_g)]$.

Theorem 9 (existence). On a nonempty compact product domain, if every R_g and resource cost c_g is continuous and the budget set is nonempty, both the minimax residual-risk program and the stated welfare-plus-disparity program attain optima.

PROOF. The budget sublevel set is closed in a compact domain, hence compact. Finite maxima, sums, and absolute differences of continuous functions are continuous. Weierstrass' theorem gives attainment. $\square$

Theorem 10 (convexity conditions). If every appeal response is concave, each cost is convex, and the domain is convex, then every R_g is convex and the minimax problem is convex in epigraph form. The separable welfare objective is convex. A disparity- penalized objective is convex only when its pairwise absolute residual-risk differences are convex (for example, under affine residual risks).

PROOF. R_g is a constant plus a nonpositive multiple of a concave response, hence convex. Finite maxima preserve convexity, and convex cost sublevel sets are convex. Weighted sums preserve convexity. The final qualification is needed because an absolute difference of two general convex functions need not be convex. $\square$

Theorem 11 (KKT marginal rule). For the differentiable convex separable-welfare program under Slater's condition, there are nonnegative multipliers η, ℓ_g, h_g satisfying

$$w_g R'_g(u_g) + \eta c'_g(u_g) - \ell_g + h_g = 0$$

and complementary slackness. If the budget binds, $0 < u_g < \bar{u}_g$, and $c'_g(u_g) > 0$, then

$$-\frac{w_g R'_g(u_g)}{c'_g(u_g)} = \eta.$$

PROOF. Under convexity and Slater's condition, KKT conditions are necessary and sufficient. Interior box constraints set $\ell_g = h_g = 0$; division by the positive marginal resource cost gives the displayed equality. Boundary groups obey inequalities rather than this equality. $\square$

Proposition 6 (minimum budget for a target). If each R_g is continuous and strictly decreasing over its effective range and each c_g is nondecreasing, define $u_g^{\min}(\epsilon) = \inf\{u : R_g(u) \leq \epsilon\}$. The common target is feasible exactly when every set is nonempty and $B \geq \sum_g c_g(u_g^{\min})$.

PROOF. Continuity makes each nonempty target sublevel closed, so the infimum is attained. Strict monotonicity makes every feasible component at least its minimum. Nondecreasing costs make the sum a necessary budget; choosing the componentwise minima proves sufficiency. $\square$

For $\text{FNR}_g > 0$ and $\rho_g > 0$, the untruncated required appeal probability is $p_g^* = (1 - \epsilon/\text{FNR}_g)/\rho_g$. If $p_g^* > 1$, equivalently $\epsilon < \text{FNR}_g(1 - \rho_g)$, the target is infeasible even with universal appeal. This boundary must not be hidden by clipping p_g^* .

Theorem 12 (discrete allocation complexity). With binary-encoded nonnegative integer intervention costs and benefits, the threshold decision version of welfare-maximizing indivisible assistance is weakly NP-complete, even for one group.

PROOF. Map an arbitrary 0–1 knapsack instance identically to individuals with the same costs and encoded expected benefits, the same budget, and the same target benefit. Feasible subsets and objective values coincide. Membership in NP follows by checking a proposed subset. The inherited hardness is weak, not strong. $\square$

A.5 Identification and statistical auditing

Theorem 13 (appeal-log non-identification). Without restrictions on appeal selection, ordinary appeal logs do not identify the final FNR even with perfect review and observed appellant labels.

PROOF. Let every case have $D_0 = 0$, fix $\mathbb{P}(A = 1) = a \in (0, 1)$ and $\mathbb{P}(Y = 1 \mid A = 1) = p \in (0, 1)$, and let perfect review set $D_1 = Y$ for appellants. Non-appellants keep $D_1 = 0$. Vary $q = \mathbb{P}(Y = 1 \mid A = 0)$ over $[0, 1]$. Since Y is unobserved exactly in the non-appellant stratum, all values of q yield the same observed law, but

$$\text{FNR}'(q) = \frac{(1-a)q}{ap + (1-a)q}$$

is strictly increasing in q . $\square$

Corollary 5 (fairness non-identification). Applying Theorem 13 separately to two groups gives observationally equivalent joint laws with different signed and absolute equal-opportunity gaps. With positive fixed appellant-positive mass, a unit gap is approached but not attained.

Theorem 14 (sharp no-assumption FNR bounds). Let $s = \mathbb{P}(D_0 = 1, Y = 1 \mid g)$, $m = \mathbb{P}(D_0 = 0, A = 1, Y = 1 \mid g)$, $n = \mathbb{P}(D_0 = 0, A = 0 \mid g)$, let ρ be the identified mean reversal probability among appealed positives, and let $z \in [0, n]$ be the latent positive mass among adverse non-appellants. Then

$$F(z) = \frac{m(1-\rho) + z}{s + m + z}.$$

When $s + m > 0$ and $s + \rho m > 0$, its sharp identified set is

$$\left[\frac{m(1-\rho)}{s+m}, \frac{m(1-\rho) + n}{s+m+n} \right].$$

If $s + m > 0$ but $s + \rho m = 0$, the set is $\{1\}$. If $s = m = 0 < n$, the set over compatible laws with a nonempty positive stratum is also $\{1\}$; if $s = m = n = 0$, the FNR is undefined under every compatible law.

PROOF. Where defined,

$$F'(z) = \frac{s + \rho m}{(s + m + z)^2} \geq 0.$$

Endpoint substitution gives the first interval. The stated boundary cases follow by direct substitution. Sharpness holds because every feasible z can be assigned inside the unobserved stratum without changing the observed law. $\square$

Proposition 7 (propensity-restricted sharp bounds). Assume $m > 0$ and $0 < \underline{\alpha} \leq \alpha \leq \bar{\alpha} \leq 1$, where $\alpha = m/(m+z)$. Then

$$z \in \left[m \frac{1-\bar{\alpha}}{\bar{\alpha}}, m \frac{1-\underline{\alpha}}{\underline{\alpha}} \right] \cap [0, n].$$

Substitution into $F(z)$ gives sharp restricted bounds.

PROOF. Solve $\alpha = m/(m+z)$ for $z = m(1-\alpha)/\alpha$. This map is decreasing on $(0, 1]$, which reverses the two propensity endpoints. Intersecting with the feasible mass interval is necessary and sufficient. If the intersection is empty, the restriction is falsified by the observed masses. $\square$

Theorem 15 (identification by representative audits). If each adverse non-appellant is audited with positive probability independently of Y conditional on the recorded stratum, and the audit reveals Y without error, then the missing prevalence q and hence $z = nq$, the final FNR, and post-contestation disparities are identified.

PROOF. Conditional independence gives

$$\mathbb{P}(Y = 1 \mid \text{audit} = 1, D_0 = 0, A = 0, G = g) = \mathbb{P}(Y = 1 \mid D_0 = 0, A = 0, G = g) = q_g.$$

All remaining terms in Theorem 14 are observed. $\square$

Proposition 8 (finite-audit contraction). For $M_g \geq 1$ independently audited non-appellants and positive fraction $\widehat{q}_g$, Hoeffding's inequality gives, with probability at least $1 - \delta$,

$$|\widehat{q}_g - q_g| \leq e_g = \sqrt{\frac{\log(2/\delta)}{2M_g}}.$$

After clipping the prevalence interval to $[0, 1]$ and propagating it through $F(n_g q)$, its width is at most

$$\frac{2n_g(s_g + \rho_g m_g)}{(s_g + m_g)^2} e_g$$

when $s_g + m_g > 0$.

PROOF. Hoeffding gives the prevalence event. Theorem 14's map is monotone, so its image is a confidence interval. Its derivative with respect to q is $n_g(s_g + \rho_g m_g)/(s_g + m_g + n_g q)^2$, bounded above by the displayed coefficient without $2e_g$. The mean-value theorem completes the bound. $\square$

Proposition 9 (simultaneous group bounds). For K groups, setting

$$e_g = \sqrt{\frac{\log(2K/\delta)}{2M_g}}$$

gives simultaneous coverage at least $1 - \delta$ for all group prevalences, FNR images, and interval-derived pairwise gaps.

PROOF. Apply Hoeffding at error probability δ/K within each group and take a union bound. Monotone image and interval arithmetic preserve the joint coverage event. $\square$

A.6 Audit disposition and code correspondence

The proof audit found no contradiction in the central signed-gap, preservation, endogenous-access, allocation, or non-identification results. It did require three substantive clarifications: Theorem 1 is now stated on a signed-perturbation space rather than inferred from an unspecified probability domain; Proposition 6 exposes infeasible targets instead of clipping the required appeal probability; and Theorem 14/Proposition 7 now handle zero-mass boundaries explicitly. Regression tests cover the new identification boundaries. The theorem-to-code correspondence remains diagnostic rather than a substitute for proof.

References

- V. Angelova, W. Dobbie, and C. S. Yang. 2023. *Algorithmic Recommendations and Human Discretion*. Tech. rep. 31747. National Bureau of Economic Research.
- M. De-Arteaga, A. Dubrawski, and A. Chouldechova. 2018. "Learning under Selective Labels in the Presence of Expert Consistency." *arXiv preprint arXiv:1807.00905*.
- B. Becker and R. Kohavi. 1996. *Adult*. UCI Machine Learning Repository. doi:[10.24432/C5XW20](https://doi.org/10.24432/C5XW20).

- J. Chen, Z. Li, and X. Mao. 2025. "Learning with Selectively Labeled Data from Multiple Decision-makers." In: *Proceedings of the 42nd International Conference on Machine Learning* (Proceedings of Machine Learning Research). Vol. 267, 8480–8519.
- J. Cobbe, M. S. A. Lee, and J. Singh. 2021. "Reviewable Automated Decision-Making: A Framework for Accountable Algorithmic Systems." In: *Proceedings of the 2021 ACM Conference on Fairness, Accountability, and Transparency*.
- A. Coston, A. Rambachan, and A. Chouldechova. 2021. "Characterizing Fairness over the Set of Good Models under Selective Labels." *arXiv preprint arXiv:2101.00352*.
- T. Freiesleben, K. Meding, and G. König. 2026. "Contestability versus Recourse: Reframing the Challenge of Contesting Algorithmic Decisions." *arXiv preprint arXiv:2605.16041*.
- T. B. Gillis, B. McLaughlin, and J. Spiess. 2022. "On the Fairness of Machine-Assisted Human Decisions." *arXiv preprint arXiv:2110.15310*.
- H. Hofmann. 1994. *Statlog (German Credit Data)*. UCI Machine Learning Repository. doi:10.24432/C5NC77.
- M. E. Kaminski and J. M. Urban. 2021. "The Right to Contest AI." *Columbia Law Review*, 121, 1957–2048.
- L. Kavouras, K. Tsopelas, G. Giannopoulos, D. Sacharidis, E. Psaroudaki, N. Theologitis, D. Rontogiannis, D. Fotakis, and I. Emiris. 2023. "Fairness Aware Counterfactuals for Subgroups." In: *Advances in Neural Information Processing Systems*. Vol. 36.
- N. Kilbertus, M. G. Rodriguez, B. Schölkopf, K. Muandet, and I. Valera. 2020. "Fair Decisions Despite Imperfect Predictions." In: *Proceedings of the Twenty Third International Conference on Artificial Intelligence and Statistics* (Proceedings of Machine Learning Research). Vol. 108.
- J. von Kügelgen, A.-H. Karimi, U. Bhatt, I. Valera, A. Weller, and B. Schölkopf. 2022. "On the Fairness of Causal Algorithmic Recourse." In: *Proceedings of the AAAI Conference on Artificial Intelligence*. Vol. 36, 9584–9594.
- Y. Liu, F. Molinari, and A. Velez. 2026. "Identification and Inference for Algorithmic Frontiers with Selective Labels." *arXiv preprint arXiv:2606.14977*.
- H. Lyons, T. Miller, and E. Velloso. 2022. "What's the Appeal? Perceptions of Review Processes for Algorithmic Decisions." In: *Proceedings of the 2022 CHI Conference on Human Factors in Computing Systems*.
- H. Lyons, E. Velloso, and T. Miller. 2021. "Conceptualising Contestability: Perspectives on Contesting Algorithmic Decisions." *Proceedings of the ACM on Human-Computer Interaction*, 5, CSCW1.
- C. Nitchals. Aug. 2026. *Pricing Appeals, Correcting Errors: A Law-and-Economics Model of Enforcement, Procedural Access, and Deterrence*. Tech. rep. SSRN, (Aug. 2026). doi:10.2139/ssrn.7295581.
- A. Perdana. 2026. "Technical Fairness and Social Legitimacy: A Conceptual Mapping of Algorithmic Fairness, Procedural Justice, and Explainable AI." *AI and Ethics*. doi:10.1007/s43681-026-01342-6.
- G. Valdrighi, I. Valera, and M. M. Raimundo. 2026. "Long-term Fairness with Selective Labels." In: *Proceedings of the 43rd International Conference on Machine Learning*. arXiv: 2605.22291.
- Z. Wang, C. Huang, K. Tang, and X. Yao. 2026. "Procedural Fairness in Machine Learning." *Journal of Artificial Intelligence Research*, 85. doi:10.1613/JAIR.1.20498.
- D. Wei. 2021. "Decision-Making under Selective Labels: Optimal Finite-Domain Policies and Beyond." In: *Proceedings of the 38th International Conference on Machine Learning* (Proceedings of Machine Learning Research). Vol. 139.
- Z. Yu, X. Xi, and P. P. Shetty. 2026. "Differential Parity: Relative Fairness Between Two Sets of Decisions." *Journal of Artificial Intelligence Research*, 86. doi:10.1613/JAIR.1.21278.

B Reproducibility Checklist for JAIR

The answers below use the checklist distributed in the official JAIR author kit downloaded on 27 September 2026.

All articles

- (1) All claims investigated in this work are clearly stated. **Yes.**
- (2) Clear explanations are given of how the work substantiates the claims. **Yes.**
- (3) Limitations or technical assumptions are stated clearly and explicitly. **Yes.**
- (4) Conceptual outlines or pseudocode descriptions of introduced AI methods and important implementation details are provided. **Yes.** The mathematical operators, optimization problems, data protocol, and executable implementation details accompany the submission.
- (5) Motivation is provided for design choices, parameters, datasets, and protocols beyond metrics. **Yes.**

Articles containing theoretical contributions

Does this paper make theoretical contributions? **Yes.**

- (1) All assumptions and restrictions are stated clearly and formally. **Yes.**

- (2) All novel claims are stated formally. **Yes.**
- (3) Proofs of all non-trivial claims are provided in sufficient detail for verification by readers with relevant expertise. **Yes.**
- (4) Complex formalism is motivated and explained clearly. **Yes.**
- (5) Mathematical notation enhances clarity and gratuitous formalism is avoided. **Yes.**
- (6) Appropriate citations are given for non-trivial theoretical tools and techniques. **Yes.**

Articles reporting computational experiments

Does this paper include computational experiments? **Yes.**

- (1) All required source code is included in an appendix or will be made publicly available upon publication. **Yes.** The complete code is publicly available at <https://github.com/Ranjan-V/JAIR-Fairness-After-the-Decision>.
- (2) The source code has a license permitting reproducibility use. **Yes.** Author-written source code is licensed under the MIT License.
- (3) The source code has a license permitting general research use. **Yes.** Author-written source code is licensed under the MIT License.
- (4) Raw, unaggregated experimental data is included or will be publicly available upon publication. **Yes.** All 820 standardized seed-level experimental output files are included in the public repository. Raw third-party UCI datasets are excluded and must be obtained from source.
- (5) The unaggregated data has a license permitting reproducibility use. **Yes.** Author-generated experimental outputs are licensed under CC BY 4.0; third-party datasets retain their original UCI source terms.
- (6) The unaggregated data has a license permitting general research use. **Yes,** under the same license distinction.
- (7) Random-number generation and seeds are described sufficiently for replication. **Yes.**
- (8) The execution environment, hardware, operating system, and software versions are described. **Partially.** The CPU-only Kaggle setting, exact dependency constraints, and clean-environment versions are recorded; the managed worker's CPU model and RAM were not exposed in the archive.
- (9) Evaluation metrics are explained and motivated. **Yes.**
- (10) The number of algorithm runs is reported. **Yes.**
- (11) Unsuccessful or unsatisfactory experiments have not been silently omitted. **Yes.**
- (12) Results include variation, confidence, or other distributional information beyond point summaries. **Yes.**
- (13) All algorithm and hyperparameter settings and their selection method are reported. **Yes.**
- (14) The range and number of settings explored before the final experiments and optimization effort are indicated. **Partially.** The predeclared final factorial and thresholds are archived; this study did not conduct a broad hyperparameter search.
- (15) Appropriate statistical hypothesis tests are used in the presence of noise. **Yes.** Paired two-sided 95% Student-*t* intervals are used for seed-level changes; conclusions are stated as interval resolutions, not as universal population claims.

Articles using datasets

Does this work rely on one or more datasets? **Yes.**

- (1) Newly introduced datasets are included or will be public with a research-use license. **Not applicable.** No new dataset is introduced.
- (2) A newly introduced dataset has a reproducibility-use license. **Not applicable.**
- (3) A newly introduced dataset has a general research-use license. **Not applicable.**
- (4) Datasets from public sources have appropriate citations. **Yes.**

- (5) Datasets from the literature are publicly available. **Yes.**
- (6) New or non-public datasets are described in detail. **Not applicable.**
- (7) Preprocessing and splitting methods are described in detail. **Yes.** The manuscript data card records source and prepared counts, label and group mappings, feature exclusion, missing-value handling, encoding, scaling, splitting, threshold calibration, models, seeds, and every semisynthetic contestation parameter; executable preprocessing is supplied.